Bank Marketing Subscription Prediction: A Machine Learning Approach

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Abstract

In this project, we worked with the Bank Marketing dataset to predict whether customers would subscribe to a term deposit. The dataset contains information about clients and the marketing campaigns of a Portuguese bank. We used three machine learning models: Support Vector Machine (SVM) and Naive Bayes, which were covered in the course, and Random Forest, which was selected for its strong performance and ability to handle complex data. The data went through cleaning, encoding, and scaling to prepare it for training. We compared the models based on accuracy, precision, recall, and F1-score. The best performing model overall was the SVM, reaching the highest accuracy. However, all models showed challenges in predicting the "yes" class due to class imbalance. Our report presents the process, results, and conclusions, with ideas for possible improvements in the future.

1. Introduction

Marketing campaigns are an important tool for banks to attract new customers and offer financial products. However, running campaigns without knowing which customers are likely to respond can waste time and resources. This project focuses on helping banks predict whether a customer will subscribe to a term deposit based on their personal and contact information, as well as past campaign outcomes.

Being able to predict customer behavior is valuable for improving marketing efficiency, saving costs, and increasing success rates. To solve this classification problem, we used two classical models, Support Vector Machine (SVM) and Naive Bayes, which are simple but effective for structured data. In addition, we used Random Forest, a more powerful ensemble model that can handle complex relationships in the data. Comparing these models helps us understand which approach works best for this kind of problem.

2. Related Work

Several recent projects have addressed the Bank Marketing dataset to solve the same classification problem. These works applied different models, preprocessing techniques, and strategies to handle the class imbalance challenge.

The first project, by Hina Ismail (Kaggle, 2023), focused on comparing several traditional machine learning models. The author applied models including Logistic Regression, Decision Tree, SVM, KNN, Random Forest, Naive Bayes, and even a simple Neural Network. The main preprocessing steps included removing the 'duration' column to avoid data leakage and encoding categorical features. The models achieved between 88–89

The second project, by Younus Mohamed (Kaggle, 2024), applied a more advanced model — the CatBoost Classifier, a gradient boosting method specifically designed to handle categorical data efficiently. This project achieved outstanding results with an AUC of 0.988 and an F1-score of 0.96, indicating a very strong ability to distinguish between classes. In addition to high predictive performance, the author also used SHAP (SHapley Additive exPlanations) to interpret model predictions and analyze which features contributed most to the decision process. This focus on interpretability provides valuable insights for understanding customer behavior.

The third project, by Peyman Daei Rezaei (Kaggle, 2023), explored classical models such as Logistic Regression, Decision Tree, Random Forest, SVM, and KNN. Like other projects, it followed best practices in preprocessing, including removing the 'duration' column and applying One-Hot Encoding to categorical variables. The models reached accuracy scores in the range of 88–89

How Our Work is Different: While previous projects like those by Hina Ismail and Peyman Daei Rezaei focused on optimizing classical models, and the project by Younus Mohamed applied an advanced boosting method (CatBoost), our project deliberately focuses on comparing simple classical models (SVM and Naive Bayes) learned in the course with a more advanced yet accessible model (Random Forest) that was not taught in class but is widely used. Unlike projects that rely heavily on advanced boosting or deep learning, our approach emphasizes a balance between simplicity, interpretability, and good predictive performance, while adhering strictly to fundamental machine learning principles taught during the course.

Furthermore, while some projects focus on squeezing maximum performance using techniques like SMOTE or complex tuning, our project aims to provide a transparent workflow that is easy to understand and reproduce, which is particularly valuable for educational purposes

3. Method

The dataset used in this project is the Bank Marketing Dataset from the UCI Machine Learning Repository. It contains data collected from direct marketing campaigns by a Portuguese bank, with the goal of predicting whether a customer will subscribe to a term deposit. The dataset includes 45,211 records and 17 features, covering client information (like age, job, marital status), financial details (like balance), contact methods, and outcomes from previous campaigns. The dataset is imbalanced, with about 88% of clients labeled as "no" (did not subscribe) and only 12% labeled as "yes" (subscribed). The dataset can be accessed at: https://archive.ics.uci.edu/ml/datasets/Bank+Marketing

Data Cleaning and Preprocessing:

- The column `duration` was removed because it represents the call duration, which is only known after the call ends and would cause data leakage.

- Categorical features were encoded using One-Hot Encoding (such as `job`, `marital`, `education`, `contact`, `month`, and `poutcome`).
- Binary categorical columns like `default`, `housing`, `loan`, and the target `y` were transformed using Label Encoding.
- Numerical features like `age`, `balance`, `day`, `campaign`, `pdays`, and `previous` were normalized using StandardScaler to improve model performance.
- The dataset had no missing values, so no imputation was needed.

Feature Selection and Engineering:

- A feature importance analysis was performed using a Random Forest model to identify the most relevant features.
- No dimensionality reduction (like PCA) was applied, since the number of features was manageable and models performed well without it.

Models Used:

- **Support Vector Machine (SVM)** — A classical supervised model that finds the optimal boundary (hyperplane) between classes. It works well for structured datasets and handles high-dimensional spaces effectively.
- **Naive Bayes** — A probabilistic model based on Bayes' theorem. It assumes that features are independent from each other, which makes it simple, fast, and often surprisingly effective for classification tasks.
- **Random Forest** — An ensemble model based on multiple decision trees. Each tree is trained on a random subset of the data and features. The final prediction is made by aggregating the results of all trees (majority voting). Random Forest is powerful because it reduces overfitting compared to a single decision tree and can handle both numerical and categorical data. Its main hyperparameters are the number of trees (`n_estimators`), tree depth, and criteria for splitting. It is also robust to noise and handles imbalanced data better than simpler models.

Final Features Used:

- All meaningful features were kept after removing `duration`. This includes client information (`age`, `job`, `marital`, `education`), financial details (`balance`, `default`, `housing`, `loan`), contact-related features (`contact`, `month`, `day`), and previous campaign outcomes (`campaign`, `pdays`, `previous`, `poutcome`). All features were encoded or scaled as needed.

4. Evaluation

To evaluate the models, the dataset was split into three parts: 70% for training, 15% for validation, and 15% for testing. This split allows us to train the models, tune parameters using the validation set, and finally check the performance on completely unseen test data. We did not use cross-validation because the data split was enough to evaluate and compare the models reliably, given the large size of the dataset.

The models were evaluated using several metrics:

- **Accuracy** – the overall percentage of correct predictions.
- **Precision** – how many of the predicted `"yes"` were actually `"yes"`.
- **Recall** – how many of the actual `"yes"` were correctly identified (especially important due to class imbalance).
- **F1-score** – the balance between precision and recall.
- **Confusion Matrix** – to visually analyze true positives, false positives, false negatives, and true negatives.

These metrics helped us compare the models not only by overall performance but also by how well they handled the minority class (`yes`), which is more challenging.

For the final Random Forest model, we selected the following hyperparameters based on experiments with the validation set:

- `n_estimators` = 100 – the number of trees in the forest.
- `max_depth` = None – trees grow until all leaves are pure or until all leaves contain less than the minimum number of samples.
- `class_weight` = None – no balancing since we decided to focus on the regular version after testing.
- `random_state` = 42 – to ensure reproducibility.

These parameters provided the best balance between accuracy and recall without overfitting.

5.Results

Table 1: Performance of the models on the validation set.

Model	Accuracy	Precision	Recall	F1-Score
SVM	**0.895**	0.67	0.20	0.31
Naive Bayes	0.852	0.38	**0.42**	0.40
Random Forest	0.885	**0.52**	0.27	**0.36**

The SVM model achieved the highest accuracy (89.5%), meaning it performed best at general classification between 'yes' and 'no'. However, it struggled significantly with the minority class ('yes'), achieving only 20% recall, which means it failed to correctly predict most customers who actually subscribed. This happens because SVM tries to create a clear boundary but tends to favor the majority class ('no') when the dataset is imbalanced.

The Naive Bayes model, although having the lowest accuracy (85.2%), showed the highest recall for the 'yes' class (42%). This means it was better at catching potential subscribers, but its precision was low (38%), so it also produced many false positives — incorrectly predicting 'yes' for customers who actually said 'no'. This reflects the nature of Naive Bayes, which assumes independence between features and can generalize well in some situations but struggles with complex relationships.

The Random Forest model offered the best balance between precision and recall. It achieved a high accuracy (88.5%), a precision of 52% (the highest among the models), and a recall of 27%, which is better than SVM but lower than Naive Bayes. Random Forest's ability to combine multiple decision trees helps it capture complex patterns and feature interactions, giving it more stable performance across both classes.

When Did Models Struggle?

SVM struggled with identifying the 'yes' class due to its strong bias toward the majority class. It prioritized correct predictions of 'no' but missed most 'yes' cases.

Naive Bayes was better at detecting 'yes' but suffered from overpredicting it, causing many false positives.

Random Forest handled the imbalance better than both, but its recall still shows the difficulty in predicting 'yes' customers due to the dominant 'no' class.

Why Are the Performances Different?

The main reason for these differences is the class imbalance (88% 'no', 12% 'yes'). Models naturally lean toward predicting the majority class unless specific techniques are used to address this (like SMOTE, class weighting, etc.).

SVM, being margin-based, works well when classes are balanced but tends to create boundaries that ignore small classes.

Naive Bayes, a probabilistic model, is less sensitive to imbalance but oversimplifies the relationships between features, which can reduce accuracy.

Random Forest, as an ensemble of trees, captures non-linear relationships and is more robust to noisy or imbalanced data but is not fully immune to imbalance problems.

6.Conclusions and Future Work

During the project, one of the main challenges was dealing with the class imbalance, where most customers did not subscribe (*no*). This made it harder for models to correctly predict the minority class (*yes*). Although Random Forest provided the best balance, the recall for the *yes* class is still not ideal.

To improve results in future work, several steps could be taken:

- Apply advanced techniques like SMOTE or other oversampling methods to improve recall for the minority class.
- Perform hyperparameter tuning (grid search or random search) to further optimize model performance.
- Try more advanced models, such as CatBoost, XGBoost, or LightGBM, which are known for handling structured data very well.
- Combine models using ensemble techniques like stacking or boosting to leverage strengths from multiple algorithms.
- Explore using more data, if available, or engineering additional features that might help the models distinguish between classes better.

Despite the challenges, the project successfully built models that can predict customer subscription with high accuracy, offering valuable insights for targeted marketing strategies.

7.Appendix

As part of this project, we used ChatGPT as a supportive tool to assist with a few tasks. The main purpose was to help improve the clarity of explanations, check correct understanding of machine learning concepts, and assist with the wording and structure of the report. Additionally, it was used to validate the logic of some coding steps and to get guidance on best practices for data preprocessing, evaluation metrics, and model comparisons.

It is important to note that all coding, model development, data analysis, and final report writing were done independently by us as students. The LLM was used only as a supportive tool to clarify doubts, polish language, and help brainstorm ideas. Every output from the LLM was carefully reviewed, modified, and adjusted by us to fit our needs and understanding.

References

[1] S. Moro, P. Cortez, and P. Rita. *A data-driven approach to predict the success of bank telemarketing.* Decision Support Systems, Volume 62, Pages 22-31, 2014. https://doi.org/10.1016/j.dss.2014.03.001

[2] Bank Marketing Dataset. UCI Machine Learning Repository. https://archive.ics.uci.edu/ml/datasets/Bank+Marketing

[3] Hina Ismail. *Bank Marketing Prediction – Using ML Models.* Kaggle Notebook, 2023. https://www.kaggle.com/code/hinaimam04/bank-marketing-prediction

[4] Younus Mohamed. *Bank Marketing ML Prediction (CatBoost Approach).* Kaggle Notebook, 2024. https://www.kaggle.com/code/younusmohamed/bank-marketing-ml-prediction-catboost

[5] Peyman Daei Rezaei. *Bank Marketing Dataset Analysis and Prediction.* Kaggle Notebook, 2023. https://www.kaggle.com/code/peymandaei/bank-marketing-analysis